

Drugreflector 药物筛选

环境搭建

创建环境

```
conda create -p /home/lizhanglong/conda_env/Drugreflector python=3.12  
  
conda activate Drugreflector
```

在克隆仓库时需要key

```
# 查看是否有SSH密钥  
ls ~/.ssh/  
# 创建  
ssh-keygen -t ed25519 -C "lizhanglongggg@163.com"  
# 查看  
cat ~/.ssh/id_ed25519.pub
```

克隆仓库

```
git clone git@github.com:Cellarity/drugreflector.git
```

安装依赖包

```
pip install torch numpy pandas scipy anndata scanpy
```

下载检查点文件

DOI: 10.5281/zenodo.16912444
下载所有检查点文件并将它们放置在checkpoints/目录中

整体运行 -- 在固定环境中在下载的文件路径下

加载包

```
import pandas as pd
import scanpy as sc
import drugreflector as dr
from utils import compute_vscores_adata
```

加载参考pbmc3k单细胞数据

```
# 加载count数据
adata = sc.datasets.pbmc3k()
# 加载metadata数据
annots = sc.datasets.pbmc3k_processed().obs
# 将metadata合并到h5ad对象中
adata.obs = pd.merge(adata.obs, annots, how='left', left_index=True,
right_index=True)
# 加载自己数据
# Single_cell = sc.read_h5ad("../Single_cell.h5ad")
```

计算V得分

```
vscores = compute_vscores_adata(
    adata,
    group_col='louvain',    # 分组参数
    group1_value='CD14+ Monocytes',    # 对照组
    group2_value='FCGR3A+ Monocytes',    # 疾病组
    layer = None
)
```

加载模型

```
# 加载模型检查点
model_paths = [
    'checkpoints/model_fold_0.pt',
    'checkpoints/model_fold_1.pt',
    'checkpoints/model_fold_2.pt'
]
model = dr.DrugReflector(checkpoint_paths=model_paths)
```

运行预测

```
# 预测前50个小分子
predictions = model.predict(vscores, n_top=50)
```

加载 Cmap 小分子数据

```
# https://clue.io/data/CMap2020#LINCS2020
Drugs = pd.read_csv('../compoundinfo_beta.txt', sep= '\t')
```

整理并合并

```
# 将索引转换成列
predictions = predictions.reset_index()
# 重命名--防止列名有多个对象
predictions.columns = ["pert_id", "rank", "logit", "prob"]
# 合并
merged = pd.merge(predictions, Drugs,
                   on= 'pert_id')
```

Seurat --> H5ad (R)

```
# 加载R包
library(Seurat)
# Biocmanager::install("zellkonverter")
library(zellkonverter)
# 加载单细胞数据
load("Single_cell.Rdata")
# 转换对象
ace <- as.SingleCellExperiment(Single_cell)
writeH5AD(ace, file = "Single_cell.h5ad")
```